Structured Gaussian Argmax Layers

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August 2026

Abstract

Softmax is the gradient of a perturbed maximum: independent Gumbel noise turns the multiclass maximum into log-sum-exp. General perturbed-optimizer expectations are typically evaluated by Monte Carlo, and the leading heteroscedastic image classifiers, whose output layer is exactly a low-rank Gaussian argmax, sample a temperature-softmax relaxation. For Gaussian perturbations with low-rank-plus-diagonal covariance, one shared conditional field yields every winner probability, the perturbed-max potential with a certified tail remainder, and, by a new reverse-mode formula, gradients with respect to means, loadings, and variances, at the asymptotic cost of a forward pass: deterministic quadrature at small factor rank, fixed-node Rao–Blackwellized factor integration in general. The dual regularizer's curvature is the effective resistance of a photo-finish graph, giving each input its own simplex geometry where softmax fixes one. The exact likelihood learns through decision-boundary rivals, and softmax is characterized as the unique link for which the Fenchel–Young loss and the likelihood coincide.

Keywords: perturbed optimizers; multinomial probit; Fenchel–Young losses; graph Laplacian; heteroscedastic classification; effective resistance

1 Introduction

A large class of neural operations is organized around the maximum of N competing scores. The maximizer is discrete, so it is smoothed by perturbation: for noise ξ with distribution determined by a covariance Σ,

$$G_\Sigma(\mu) = \mathbb{E}\Big[\max_i(\mu_i + \xi_i)\Big], \qquad p_\Sigma(\mu) = \nabla_\mu G_\Sigma(\mu) = \mathbb{E}\big[e_{\arg\max_i(\mu_i+\xi_i)}\big].$$

For independent Gumbel noise G is log-sum-exp and p is softmax, up to an additive constant. Berthet et al. (2020) develop the general perturbed-optimizer view, including the convex dual and the induced Fenchel–Young loss (Blondel et al., 2020); Indelman and Hazan (2024) study the representation properties of perturb-softmax and perturb-argmax families, including Gaussian members; and Lin et al. (2026) develop Fenchel–Young estimation for perturbed-utility models, including the logit case where it coincides with maximum likelihood. In all of these, general perturbed expectations are typically evaluated by Monte Carlo.

The structured Gaussian member is the natural carrier of correlation. We study

$$U_i = \mu_i + v_i^\top f + \sqrt{D_i}\,\epsilon_i, \qquad f \sim N(0, I_k), \quad \epsilon_i \sim N(0,1) \text{ iid},$$

*Working draft. Every identity below is numerically checked (experiment 27 of the companion repository, `https://github.com/microprediction/kinetics`); the neural experiments of Section 8 are pending.

so $\Sigma = VV^\top + \mathrm{diag}(D)$. The structure arises automatically from uncertain representations: if a feature vector is $h = \bar{h} + Af$ and scores are $U_i = w_i^\top h + b_i + \sqrt{D_i}\,\epsilon_i$, then $v_i = A^\top w_i$.

The same model already ships in computer vision. Collier et al. (2021) place exactly this latent-utility argmax, with input-dependent low-rank-plus-diagonal covariance, on the final layer of ImageNet-scale classifiers; lacking a closed form, they approximate the winner probabilities by Monte Carlo over a temperature-softmax relaxation, and the scaled follow-up (Collier et al., 2023) identifies the latency and memory cost of that sampling as a standing issue.

A numerical companion (Cotton, 2026) supplies the missing primitive: conditional on the factors the alternatives are independent, so all N winner probabilities come from a single product field in $O(QNL)$ operations for Q factor nodes and L lattice points, with exact Jacobian–vector products, a weighted graph-Laplacian Jacobian, and a globally invertible map from mean utilities modulo translation to the simplex interior.

This paper develops that primitive as a neural layer. The novelty boundary is narrow: perturbed-max smoothing, duality, Fenchel–Young losses, and the Gaussian-argmax family as such are prior art (Berthet et al., 2020; Blondel et al., 2020; Indelman and Hazan, 2024; Lin et al., 2026; Fosgerau et al., 2020). What we add is the structured-Gaussian computational and geometric realization:

1. **The potential is a same-field reduction**, with a certified closed-form tail remainder (Section 3).

2. **Full shared-field reverse mode**: gradients in (μ, V, D) at the same asymptotic complexity as a forward pass (Section 5), with a cheaper label-conditioned kernel for supervised likelihood training.

3. **Input-dependent simplex geometry**: the dual regularizer's curvature is effective resistance in the photo-finish graph, so a heteroscedastic model $x \mapsto \Sigma(x)$ learns a geometry per input where softmax fixes the Shannon geometry for every input (Section 4).

4. **Boundary-aware likelihood learning**: the exact winner likelihood performs model-implied hard-negative mining through the label's photo-finish edges; its linear-head problem is convex; and softmax is characterized as the unique smooth link whose Fenchel–Young one-hot loss is its likelihood (Section 6).

5. **What the surrogate computes**: the Monte-Carlo temperature-softmax construction is the exact winner probability of a Gaussian-plus-Gumbel model, so temperature selects a noise family rather than merely trading bias for variance (Section 7).

6. **A scale-collapse caution**: the Fenchel–Young objective is monotone in choice-relevant perturbation scale, so unconstrained covariance-scale learning collapses; shape learning requires a constraint, a bilevel criterion (Lin et al., 2026), or the likelihood (Section 6).

Every displayed identity is numerically checked (Table 1); the neural experiments of Section 8 are pending.

2 The structured Gaussian argmax

Conditional on f, the scores are independent with means $m_i(f) = \mu_i + v_i^\top f$ and variances D_i. Write $F_i(x \mid f)$ and $g_i(x \mid f)$ for the conditional CDF and density. Then

$$p_i = \mathbb{P}(U_i = \max_j U_j) = \mathbb{E}_f \int g_i(x \mid f) \prod_{j \neq i} F_j(x \mid f)\, dx.$$

The shared field is $H_f(x) = \prod_j F_j(x \mid f)$; each winner integrand divides one factor back out. On Q factor nodes and an L-point lattice, all N probabilities cost $O(QNL)$ (Cotton, 2026).

Three integration regimes. At small k the factor expectation is deterministic product Gauss–Hermite quadrature. At moderate k it is fixed scrambled-Sobol quasi–Monte Carlo: reproducible and resampling-free, but an integration rule, not a deterministic quadrature. At larger k it is factor-only (Q)MC, and this is a Rao–Blackwellization: with $q_i(f) = \mathbb{E}[\mathbf{1}\{i = \arg\max\} \mid f]$ computed by the field, $p_i = \mathbb{E}_f q_i(f)$ and, by total variance,

$$\mathrm{Var}[q_i(f)] \leq p_i(1 - p_i):$$

sampling only f while integrating all N idiosyncratic shocks analytically can only reduce variance relative to direct noisy-argmax simulation. How much it reduces it, as a function of the factor share of variance, is an empirical question of Section 8.

3 The potential from the same field

Conditional on f, H_f is the CDF of the maximum M_f, so for any finite anchor a,

$$\mathbb{E}[M_f \mid f] = a + \int_a^\infty [1 - H_f(x)]\, dx - \int_{-\infty}^a H_f(x)\, dx,$$

and $G_\Sigma(\mu)$ is the factor average. Taking a at the lattice endpoint b and dropping the upper tail leaves the computable truncation

$$\mathbb{E}[M_f \mid f] = b - \int_{-\infty}^b H_f(x)\, dx + R_b(f), \qquad R_b(f) = \int_b^\infty [1 - H_f(x)]\, dx \geq 0,$$

with the certified closed-form bound

$$R_b(f) \leq \sum_i \mathbb{E}[(U_i - b)_+ \mid f] = \sum_i \Big[\sqrt{D_i}\, \phi(a_i) - (b - m_i(f))\, \bar{\Phi}(a_i)\Big], \qquad a_i = \frac{b - m_i(f)}{\sqrt{D_i}}.$$

At the implemented 8σ endpoint the measured remainder is 10^{-25} against a bound of 2.3×10^{-22} (Table 1): the truncation is a controlled approximation, not an identity, and the bound makes it certified. The forward pass has already built H_f at every node and lattice point; the potential is one more reduction.

Which map is differentiated? Four maps must be distinguished: the exact continuum p_Σ; the fixed-node, fixed-grid quadrature map $\tilde{p}_{Q,L}$; its normalization $p_{Q,L} = \tilde{p}_{Q,L}/\mathbf{1}^\top \tilde{p}_{Q,L}$; and an adaptive-envelope implementation whose lattice endpoints move with the parameters. The reverse-mode formulas below are exact for the continuum integrals, and exact for the fixed-grid sums once the normalization cotangent is pulled back; they are not automatically the derivative of the adaptive map unless endpoint motion is differentiated too (the truncated potential's boundary term $\partial_\theta b\,[1 - H_f(b)]$, which the full anchor identity cancels). For Fenchel–Young training the clean construction makes $G_{Q,L}$ the primary numerical object and *defines* $p_{Q,L} = \nabla_\mu G_{Q,L}$ as its exact discrete gradient: integrability then holds by construction, and building translation equivariance into $G_{Q,L}$ gives $\mathbf{1}^\top p_{Q,L} = 1$ automatically. An independently discretized pair broke $\nabla G = p$ at 2.5×10^{-4} in our tests; the consistent pair restores 3.6×10^{-8}.

4 Dual geometry

G_Σ is convex and $G_\Sigma(\mu + c\mathbf{1}) = G_\Sigma(\mu) + c$, so the conjugate $\Psi_\Sigma = G^*_\Sigma$ lives on the simplex and, on the mean-zero gauge, $p = \nabla G_\Sigma(\mu)$ and $\mu = \nabla \Psi_\Sigma(p)$: the share inversion of the companion paper is the mirror map, and $\Psi_\Sigma(p) = p^\top \mu(p) - G_\Sigma(\mu(p))$ is computable from the inverse transform plus the same-field potential.

The Hessian is the photo-finish Laplacian

$$J_\Sigma = \nabla^2_\mu G_\Sigma = \sum_{i<j} w_{ij}(e_i - e_j)(e_i - e_j)^\top, \qquad w_{ij} = \mathbb{E}_f \int g_i g_j \prod_{\ell \neq i,j} F_\ell \, dx > 0,$$

with w_{ij} the density of an i–j tie for the winning position. On the tangent space, $\nabla^2 \Psi_\Sigma(p) = J^+_\Sigma$, and for $d_{ij} = e_i - e_j$,

$$d_{ij}^\top J^+_\Sigma d_{ij} = \text{effective resistance between } i \text{ and } j$$

in the network with conductances w_{ab}: probability moves cheaply between strongly competing alternatives and expensively across weak competitive pathways.

Temperature-τ softmax is the special case $w_{ij}^{\text{soft},\tau} = p_i p_j/\tau$, whose resistance is $\tau(1/p_i + 1/p_j)$ — the curvature of τ-scaled negative Shannon entropy. Shannon geometry is the effective-resistance geometry of one particular competition graph.

At fixed Σ, the companion inversion theorem says p determines μ modulo translation, hence also $J_\Sigma(\mu)$: the graph is not extra information beyond p *within* one geometry. Across covariances it is: the same p carries different photo-finish Laplacians under different Σ. That is the point for heteroscedastic classification. An input-dependent model $x \mapsto \Sigma(x)$ induces $x \mapsto \Psi_{\Sigma(x)}$: each input gets its own simplex geometry, distinguishing "similar marginal probability" from "direct competitors at this input's decision boundary," where softmax uses the same Shannon geometry for every input.

5 Shared-field reverse mode

The companion paper differentiates p in μ. A heteroscedastic layer needs V and D as well. Let $r_i(x, f) = g_i \prod_{\ell \neq i} F_\ell$ be the winner densities, let a downstream loss supply the cotangent $c_i = \partial \mathcal{L}/\partial p_i$, and form one additional field $S_c = \sum_i c_i r_i$ with hazards $\lambda_j = g_j/F_j$. Define

$$A_j(x, f) = c_j r_j \frac{x - m_j}{D_j} - (S_c - c_j r_j)\,\lambda_j, \qquad B_j(x, f) = c_j r_j \frac{(x - m_j)^2 - D_j}{2D_j^2} - (S_c - c_j r_j)\,\frac{x - m_j}{2D_j}\,\lambda_j.$$

Proposition 1 (Shared-field vector–Jacobian product)**.**

$$\frac{\partial \mathcal{L}}{\partial \mu_j} = \mathbb{E}_f \int A_j \, dx, \qquad \frac{\partial \mathcal{L}}{\partial v_j} = \mathbb{E}_f \Big[f \int A_j \, dx \Big], \qquad \frac{\partial \mathcal{L}}{\partial D_j} = \mathbb{E}_f \int B_j \, dx.$$

All N parameter blocks reuse the same S_c; the cost is $O(QN(L+k))$, the same asymptotic complexity as a forward pass.

Proof. Only two terms depend on alternative j's parameters: its own winner density through g_j, and every other winner density through F_j. Hence $\partial_{\theta_j} \mathcal{L} = \mathbb{E}_f \int [c_j r_j \, \partial_{\theta_j} \log g_j + (S_c - c_j r_j) \, \partial_{\theta_j} \log F_j] \, dx$. For a Gaussian factor, $\partial_{m_j} \log g_j = (x - m_j)/D_j$ and $\partial_{m_j} \log F_j = -\lambda_j$; $\partial_{D_j} \log g_j = ((x-m_j)^2 - D_j)/(2D_j^2)$ and $\partial_{D_j} \log F_j = -\lambda_j (x - m_j)/(2D_j)$. Finally $\partial_{\mu_j} m_j = 1$ and $\partial_{v_j} m_j = f$. □

Measured against central finite differences, the three blocks agree to 2.2×10^{-9}, 4.9×10^{-10}, and 6.5×10^{-10}.

If the implementation normalizes, $p = \tilde{p}/s$ with $s = \mathbf{1}^\top \tilde{p}$, the incoming cotangent must first be pulled back: $\tilde{c} = (c - (c^\top p)\mathbf{1})/s$. Differentiating the unnormalized field while treating normalization as constant is incorrect.

The label-conditioned kernel. Supervised likelihood training has the sparse cotangent $c = -e_y/p_y$, so $S_c = -r_y/p_y$: the backward pass needs only the common field, the label's winner density r_y, and the hazards. In particular the off-label mean gradients collapse to $\partial \mathcal{L}_{\mathrm{NLL}}/\partial \mu_j = w_{yj}/p_y$. A dedicated kernel exploiting this has the same asymptotics but materially lower constants and memory, and admits a stable formulation in the normalized winning-score density $\pi_y = r_y/p_y$, under which gradient components are expectations rather than ratios of small numbers.

6 Learning objectives

Fenchel–Young. For a one-hot label e_y and centered Gaussian noise, $\Psi_\Sigma(e_y) = 0$, so the Fenchel–Young loss induced by G_Σ is

$$L_{\mathrm{FY}}(\mu, y) = G_\Sigma(\mu) - \mu_y, \qquad \nabla_\mu L_{\mathrm{FY}} = p - e_y,$$

the Gaussian Fenchel–Young counterpart of the softmax cross-entropy gradient. In the binary case, with $\delta = \mu_1 - \mu_2$ and $s^2 = \mathrm{Var}(\xi_1 - \xi_2)$, $L_{\mathrm{FY}} = s[\phi(\delta/s) - (\delta/s)\Phi(-\delta/s)] = \mathbb{E}[(U_2 - U_1)_+]$: a Gaussian-smoothed hinge whose curvature, and whose residual gradient on correctly separated examples, concentrate near photo finishes (checked to 4.9×10^{-13} against the field computation).

Likelihood as hard-negative mining. Treating the race as a probabilistic model gives $L_{\mathrm{NLL}} = -\log p_y$, and by symmetry of J_Σ, with $d_y = \sum_{j \neq y} w_{yj} = (J_\Sigma)_{yy}$ and rival distribution $\rho_{j|y} = w_{yj}/d_y$,

$$\nabla_\mu L_{\mathrm{NLL}} = -\frac{J_\Sigma e_y}{p_y} = \frac{d_y}{p_y} \left(\rho_{\cdot|y} - e_y \right) :$$

the update pushes the label's utility up and redistributes exactly onto its *decision-boundary rivals*, with total strength d_y/p_y. For a linear head $\mu_i = a_i^\top h$, $\nabla_h L_{\mathrm{NLL}} = \frac{1}{p_y} \sum_{j \neq y} w_{yj} (a_j -$

a_y): the representation moves specifically away from the class directions that share winning boundaries with y. Softmax forces $\rho_{j|y} = p_j/(1-p_y)$; the structured Gaussian model lets a class with moderate marginal probability be a dominant rival, or a probable class be irrelevant to this particular boundary. The exact likelihood performs model-implied hard-negative mining.

Proposition 2 (Convexity of the winner likelihood)**.** *For fixed Σ with $D_i > 0$, $p_y(\mu)$ is log-concave in μ, so $-\log p_y(\mu)$ is convex; with fixed features and covariance the linear-head likelihood problem is convex.*

Proof. $p_y(\mu) = \int_{C_y} \phi_\Sigma(u-\mu)\,du$ with $C_y = \{u : u_y \geq u_j\ \forall j\}$ a convex cone. The integrand $\mathbf{1}_{C_y}(u)\,\phi_\Sigma(u-\mu)$ is jointly log-concave in (u,μ); marginalizing u preserves log-concavity by Prékopa's theorem. □

Numerically, second differences of $-\log p_y$ along random directions are bounded below by $+3 \times 10^{-2}$ in our checks (Table 1). General perturbed-utility maximum likelihood need not be convex, a motivation Lin et al. (2026) give for Fenchel–Young estimation; the Gaussian winner model has more structure.

Proposition 3 (Softmax is the coincidence case)**.** *Suppose a smooth perturbed-max link $p = \nabla G$ satisfies, for all μ and every label y, $G(\mu) - \mu_y = \tau\,[-\log p_y(\mu)] + c_y$ for constants $\tau > 0, c_y$. Then p is temperature-τ softmax with class offsets: $p_y(\mu) \propto A_y e^{\mu_y/\tau}$, $A_y = e^{c_y/\tau}$. Conversely these links satisfy the identity.*

Proof. Rearranging, $p_y(\mu) = A_y \exp[(\mu_y - G(\mu))/\tau]$. Summing over y and using $\mathbf{1}^\top p = 1$ gives $G(\mu) = \tau \log \sum_y A_y e^{\mu_y/\tau}$, hence the softmax form. The converse is the standard log-sum-exp computation (checked to machine precision). □

Up to temperature and class offsets, softmax is the unique smooth perturbed-max link whose Fenchel–Young one-hot loss is also its negative log likelihood. Under any other Gaussian geometry the two objectives, and their curvatures, separate: Fenchel–Young curvature is J_Σ, while the *expected* NLL curvature is $J_\Sigma \operatorname{diag}(p)^{-1} J_\Sigma$, and softmax satisfies $J \operatorname{diag}(p)^{-1} J = J$.

Scale collapse. The Gaussian potential obeys the heat identity $dG_\Sigma = \frac{1}{2}\operatorname{tr}(J_\Sigma\, d\Sigma)$, hence

$$\frac{\partial G}{\partial D_i} = \frac{(J_\Sigma)_{ii}}{2} > 0, \qquad \nabla_V G = J_\Sigma V$$

(both checked to $\sim 10^{-8}$). Since $-\mu_y$ carries no covariance dependence, the Fenchel–Young objective is monotone increasing in every positive-semidefinite direction of choice-relevant perturbation covariance: *unconstrained learning of Gaussian perturbation scale collapses.* This does not preclude learning covariance *shape* at fixed scale, where directions are indefinite; shape learning requires a scale constraint (for instance $\operatorname{tr}(P\Sigma P)/(N-1) = 1$), a bilevel criterion (Lin et al., 2026), or the likelihood objective.

Gauges. An end-to-end layer must remove three invariances: $\mu \mapsto \mu + c\mathbf{1}$; $V \mapsto V + \mathbf{1}a^\top$; and joint scaling $(\mu, V, D) \mapsto (s\mu, sV, s^2 D)$. Project μ and V onto mean-zero columns; then compute $s^2 = \operatorname{tr}(P\Sigma P)/(N-1)$ and rescale *all three* components, $\mu \leftarrow \mu/s$, $V \leftarrow V/s$, $D \leftarrow D/s^2$, which leaves the winner probabilities unchanged. Rescaling only (V, D) changes the model rather than fixing its gauge. Factor rotation $V \mapsto VR$ remains unidentified and is harmless for prediction.

identity	metric	discrepancy
$\nabla_\mu G = p$	max abs, vs FD	3.6×10^{-8}
reverse mode, μ block	max abs, vs FD	2.2×10^{-9}
reverse mode, V block	max abs, vs FD	4.9×10^{-10}
reverse mode, D block	max abs, vs FD	6.5×10^{-10}
heat identity $\partial G/\partial D_i = J_{ii}/2$	max abs, vs FD	1.0×10^{-8}
$\nabla_V G = JV$	max abs, vs FD	7.7×10^{-9}
$\nabla_\mu(-\log p_y) = -Je_y/p_y$	max abs, vs FD	5.1×10^{-9}
binary Fenchel–Young closed form	abs	4.9×10^{-13}
dual curvature = effective resistance	relative, vs FD of Ψ	1.4×10^{-6}
softmax $R_{ij} = 1/p_i + 1/p_j$	abs, linear algebra	1.8×10^{-15}
G vs 4×10^6-draw MC	abs ($\frac{1}{2}$ s.e.)	1.9×10^{-4}
tail remainder vs certified bound	$10^{-25} \le 2.3 \times 10^{-22}$	holds
surrogate = Gauss+Gumbel argmax	max abs, vs MC (3 s.e. 10^{-3})	4.4×10^{-4}
NLL convexity, 20 random directions	min second difference	$+3.0 \times 10^{-2}$
softmax FY = NLL	abs	4.4×10^{-16}
active-class sandwich and w_{yj} bound	containment	holds

Table 1: Principal identities, numerically checked (experiment 27: $N = 5$, $k = 1$, Gauss–Hermite 21, $L = 2001$; frozen grid; central differences with steps 10^{-4}–10^{-6}). A numerical experiment checks an implementation; randomized multi-problem sweeps and autograd `gradcheck` are part of the experimental program.

7 What the temperature-softmax surrogate computes

The Monte-Carlo temperature-softmax construction of Collier et al. (2021) is not merely a smoothed estimate of the Gaussian argmax. By the Gumbel-max identity, for iid standard Gumbels γ_j,

$$\mathrm{softmax}_\tau(\mu + \xi)_i = \mathbb{P}\big(i = \arg\max_j \{\mu_j + \xi_j + \tau\gamma_j\} \,\big|\, \xi\big),$$

so averaging over the Gaussian ξ gives *exactly* the winner probability of a Gaussian-plus-Gumbel random-utility model (checked within Monte Carlo error, Table 1). The surrogate at temperature τ is the exact probability of a different noise family: $\tau = 0$ is pure Gaussian argmax, $\tau > 0$ adds iid Gumbel. Tuning τ selects a noise model, not merely a bias–variance point. The convolved model retains conditional independence given f, so it sits inside the shared-field architecture with a tabulated one-dimensional CDF; the exact-vs-surrogate comparison of Section 8 is therefore a comparison of noise families computed by one method, not of two numerical methods.

Certified screening for large N. For a rival subset S, $p_y^{(S)} = \mathbb{P}(U_y > U_j\ \forall j \in S)$ satisfies

$$\max\{0,\ p_y^{(S)} - \delta_S\} \le p_y \le p_y^{(S)}, \qquad \delta_S = \sum_{j \notin S \cup \{y\}} \Phi\Big(\frac{\mu_j - \mu_y}{s_{jy}}\Big), \quad s_{jy}^2 = D_j + D_y + \|v_j - v_y\|^2,$$

giving a certified likelihood interval whenever $p_y^{(S)} > \delta_S$; and the edge weights obey $w_{yj} \le \phi\big((\mu_y - \mu_j)/s_{jy}\big)/s_{jy}$, screening gradient contributions. Both bounds are checked numerically. This is the route to very large class sets.

8 Experimental program

Three questions, kept separate. **A** (numerical): can the operator be evaluated and differentiated accurately without resampling? **B** (optimization): does a fixed accurate gradient beat M-sample Monte Carlo gradients at matched wall time — including $M = 1$, which Berthet et al. (2020) note is often adequate for SGD? **C** (statistical): does structured Gaussian competition improve the model? A faster operator is not evidence of a better inductive bias, and vice versa.

The planned sequence: (I) continuum/discretization tests, making the numerical loss–gradient pair exactly consistent (Section 3); (II) the Rao–Blackwell benchmark — direct argmax MC, Gaussian-draw temperature softmax, factor-only MC and factor-only Sobol over $q_i(f)$, and low-rank product quadrature, across $N \in \{10, \dots, 10^4\}$ and $k \leq 15$, at equal wall time and equal error; (III) the label-conditioned NLL kernel against the generic VJP, for speed and memory; (IV) frozen-feature linear heads (convex by Proposition 2): softmax cross-entropy, Gaussian Fenchel–Young at $M \in \{1, \dots, 1000\}$, the fixed-node evaluator, and exact winner NLL, on objective gap and accuracy versus wall clock; (V) replication of the Gaussian CIFAR-10 experiment of Berthet et al. (2020) plus one treatment, the fixed-node evaluator; (VI) the heteroscedastic classifier of Collier et al. (2021), comparing pure Gaussian argmax NLL against the Gaussian-plus-Gumbel model that its temperature-softmax training actually implements (Section 7), both evaluated under a common high-accuracy reference; (VII) certified active-class screening at large N.

Success requires any of: materially better accuracy per wall time than Monte Carlo; materially lower gradient error at matched time; faster convergence to a validation threshold; large latency or memory savings at high accuracy; or better-calibrated models from the exact likelihood. The project pivots if $M = 1$ Monte Carlo trains as well and faster, if useful ranks are unreachable, or if structure adds no statistical gain — in which case the operator's role is high-accuracy inference and calibration, not SGD replacement.

9 Discussion

Softmax is the explicitly computable Gumbel member of the perturbed-max family. The results above make the structured Gaussian member explicit too: potential, probabilities, and parameter gradients from one shared-field pass, with the inverse obtained by repeated shared-field evaluations, and a graph geometry that softmax collapses to a single special case. The claim is not that Gaussian perturbations dominate Gumbel ones — indeed Section 7 shows the practical surrogate interpolates between them — but that in this structured regime, computability no longer decides the comparison in advance. The heteroscedastic classification literature already states the demand: its generative model is this model, and its current evaluator is the exact probability of a different noise family.

Reproducibility

Experiment 27 in the companion repository implements the operator and checks every identity in Table 1; the companion paper's experiments 13–26 cover the underlying transform, its accuracy assessment, and its benchmarks.

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